

Aayush Language Teaching Guide

A simple Nepali chat-style programming language for first lessons, demos, and practice.

1. What Students Should Learn First

Aayush is designed to feel close to everyday Nepali-style speech while still teaching real programming ideas: variables, output, conditions, loops, functions, and input.

- A program runs from top to bottom.
- Variables store values with `maan`, `maana_ki`, `rakha`, or `let`.
- Blocks begin with words like `yedi ... bhane` and end with `sakiyo`.
- Indentation is for readability, and the VS Code extension now helps with it.

2. First Program

Start with output. The easiest command is `lekh` or `lekha`, which prints text.

```
lekh "Namaste Aayush"

maan naam chai "Aayush"
lekh "Namaste " + naam
```

3. Variables and Values

Use `chai` as the Aayush-style assignment word. The normal equals sign also works.

Idea	Aayush example	Meaning
Text	<code>maan naam chai "Aayush"</code>	Store a string
Number	<code>maan age chai 18</code>	Store a number
Update	<code>maan age chai age + 1</code>	Replace with a new value
Print	<code>lekh "Age: " + age</code>	Show the value

4. Operators

Aayush supports normal math and comparison symbols. It also supports two friendly word aliases.

Operator	Aayush word	Use
<code>=</code>	<code>chai</code>	assignment
<code>==</code>	<code>barabar</code>	equal comparison
<code>!=</code>		not equal
<code>></code> , <code><</code>		greater than, less than

Operator	Aayush word	Use
>=, <=		greater/less than or equal
+, -, *, /		math

5. If / Else Blocks

Use yedi for if, bhane or chabhane for then, natra for else, and sakiyo to end the block.

```
maan age chai 18

yedi age >= 18 chabhane
    lekh "Timi adult hau"
natra
    lekh "Timi ajhai sano chau"
sakiyo
```

Teaching tip: read it aloud as, 'If age is greater than or equal to 18, then print adult; otherwise print small.'

6. Loops

Use jaba_samma for while and gara for do. The loop repeats while the condition is true.

```
maan count chai 1

jaba_samma count <= 5 gara
    lekh "Round " + count
    maan count chai count + 1
sakiyo
```

7. Input and Number Conversion

sodha asks the user for text. ank converts text into a number so math works correctly.

```
maan first chai ank(sodha("First number: "))
maan second chai ank(sodha("Second number: "))

lekh "Total: " + (first + second)
```

8. Functions

Functions let students name a reusable action. Use kaam for def and firta for return.

```
kaam jod(a, b)
    firta a + b
sakiyo

lekh jod(4, 5)
```

9. Keyword Cheat Sheet

English	Aayush	Example
let	maan / maana_ki / rakha	maan x chai 10
print	lekh / lekha / bol	lekh "Hi"
if	yedi	yedi x > 5 chabhane
then	bhane / chabhane	condition chabhane
else	natra	natra
end	sakiyo	sakiyo
while	jaba_samma	jaba_samma x < 5 gara
do	gara	loop condition gara
function	kaam	kaam add(a, b)
return	firta	firta a + b
true / false	sahi / galat	maan ok chai sahi
and / or / not	ra / wa / haina	x > 1 ra x < 10

10. Common Mistakes to Teach Early

- Every yedi, jaba_samma, and kaam block needs sakiyo.
- Use barabar for comparison, not chai. Example: yedi choice barabar 1 chabhane.
- Use ank(sodha(...)) when the answer must be a number.

- Keep nested blocks indented so students can see what belongs inside what.

11. Mini Lesson Plan

Time	Topic	Activity
10 min	Print and strings	Students print their name.
15 min	Variables	Store name and age, then print a sentence.
20 min	If / else	Check adult/minor or pass/fail.
20 min	Loops	Print numbers from 1 to 5.
25 min	Input	Build a tiny calculator.
15 min	Functions	Create jod(a, b) and use it twice.

12. Practice Exercises

- Write a program that asks for a name and prints Namaste plus the name.
- Ask for marks. If marks are 40 or more, print Pass; otherwise print Try again.
- Print numbers from 1 to 10 with jaba_samma.
- Make a function called guna(a, b) that returns a * b.
- Challenge: build a calculator with choices for +, -, *, and /.

Teacher Script: Explaining Blocks

When teaching yedi/natra/sakiyo, draw a box around the indented lines. Explain that the box is the part controlled by the condition.

```
yedi condition chabhane
    # this line is inside the if block
natra
    # this line is inside the else block
sakiyo
```

The words yedi, natra, and sakiyo show the structure. The spaces at the start of each line help our eyes understand the structure.

Final Demo Program

```
lekh "Aayush calculator"

maan first chai ank(sodha("First number: "))
maan second chai ank(sodha("Second number: "))
maan choice chai sodha("Choose +, -, *, /: ")

yedi choice barabar "+" chabhane
    lekh "Answer: " + (first + second)
natra
    yedi choice barabar "-" chabhane
        lekh "Answer: " + (first - second)
    natra
        yedi choice barabar "*" chabhane
            lekh "Answer: " + (first * second)
        natra
            yedi second barabar 0 chabhane
                lekh "Cannot divide by zero"
            natra
                lekh "Answer: " + (first / second)
            sakiyo
        sakiyo
    sakiyo
sakiyo
```